

Curriculum Vitae

PERSONAL INFORMATION

Daniel Douglas
Postdoctoral Researcher
Boston College
Earth Sciences
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EDUCATION

- 08/2021–08/2025 **Ph.D., Geophysics**
New Mexico Institute of Mining and Technology, Socorro, NM, USA
Thesis: Development and Application of Geodynamic Software for Simulating Globally Coupled 3-Dimensional Deformation and Reactive Fluid Transport at Subduction Zones
Advisor: Dr. John Naliboff
Course Work: Subduction System Processes, Theoretical Seismology, Graduate Seminar, Statistical Methods in Earth Sciences
- 08/2019–08/2021 **M.Sc., Geophysics**
University of Hawaii at Manoa, Honolulu, HI, USA
Thesis: Magma-Assisted Flexure of Hawaiian Lithosphere Inferred From Three-Dimensional Models of Lithospheric Flexure and Active Source Seismic Data
Advisor: Dr. Garrett Apuzen-Ito
Course Work: Plate Tectonics, Geophysics I, Geophysics II, Submarine Volcanology, Mineralogy
- 09/2015–08/2019 **B.Sc., Physics and Astrophysics**
University of Toronto, Toronto, Canada
Thesis: Simulating Convection in a 3D Spherical Shell
Advisor: Dr. Julian Lowman
Course Work: Differential equations, Multivariable Calculus, Continuum Mechanics, Classical Mechanics, Electromagnetism, Quantum Mechanics

PROFESSIONAL EMPLOYMENT

- 2025–Present Position: ASPECT Principal Developer
- 2024–Present Position: Geodynamic World Builder Principal Developer
- 2025–Present **Boston College**
Position: Postdoctoral Researcher
- 2021–2025 **New Mexico Institute of Mining and Technology**
Position: Research Assistant

2022–2023	USGS-NSF Intern Topic: Constraining Permeability of Subducting Lithosphere at the Central American Trench Using PyLith Sponsor: Dr. Brad Aagaard
2019–2021	University of Hawaii at Manoa Position: Research Assistant
2018–2019	University of Toronto Position: Teaching Assistant Class: Introduction to Physics I and II
2017–2019	University of Toronto Position: Facilitated Study Group Leader Class: Introduction to Physics I and II

PUBLICATIONS

Journal Publications

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| 2024 | Douglas, D. , Apuzen-Ito, G., Boston, B., Dunn, R., Naliboff, J., Wessel, P., Watts, A. B., Shillington, D., Cilli, P., Magma-Assisted Flexure of Hawaiian Lithosphere Inferred From Three-Dimensional Models of Lithospheric Flexure Constrained by Active Source Seismic Data,
https://doi.org/10.1029/2024JB030376 |
| 2024 | Douglas, D. , Naliboff, J., Fraters, M., Dannberg, J., Eberhart-Phillips, D., Ellis S., Constraining Solid Dynamics, Interface Rheology, and Slab Hydration in the Hikurangi Subduction Zone Using 3D Fully Dynamic Models,
https://doi.org/10.1029/2024GC011824 |
| 2024 | Fraters, M., Billen, M., Gassmoeller, R., Saxena, A., Heister, T., Li, H., Douglas, D. , Dannberg, J., Bangerth, W., Wang, Y., (2024). The Geodynamic World Builder: A planetary structure creator for the geosciences. <i>Journal of Open Source Software</i> , 9(101), 6671, https://doi.org/10.21105/joss.06671 |

Other Publications

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| 2021 | Douglas, D. , Magma-Assisted Flexure in Hawaiian Lithosphere? <i>Blog of the Geodynamics Division of the European Geosciences Union</i> ,
https://blogs.egu.eu/divisions/gd/2021/11/17/magma-assisted-flexure-in-hawaiian-lithosphere/ |
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RESEARCH TALKS

1. **Investigating the Role of Fluid–Solid Coupling on Subduction Dynamics and Fluid Pathways**, MPG/SGT Seminar, Lamont-Doherty Earth Observatory 2026
2. **Investigating the Role of Fluid–Solid Coupling on Subduction Dynamics and Fluid Pathways**, EES Seminar, Colorado University, Boulder 2025
3. **Constraining Permeability of Subducting Lithosphere at the Central American Trench**, Oral presentation at AGU 2025 Fall Meeting in New Orleans, LA, USA
4. **The Role of Solid–Fluid Coupling on 2-Dimensional Solid Dynamics and Fluid Migration**, Invited Seminar at Colorado State University, 2025

5. **3D Modeling of Dynamics in the Hikurangi-Kermadec Subduction System**, Oral presentation at AGU 2023 Fall Meeting in San Francisco, CA, USA

CONFERENCES AND WORKSHOPS

1. **12th ASPECT Hackathon**, 2025, Garden City, UT, USA
2. **2nd WorldBuilder Hackathon**, 2025, Potsdam, Germany
3. **11th ASPECT Hackathon**, 2024, CSU Mountain Campus, CO, USA
4. **1st WorldBuilder Hackathon**, 2024, Panama City, FL, USA
5. **Geodynamic Constraints on the Stability of Forearc Serpentinite in the Cold Nose of the Mantle Wedge**, Poster presentation at AGU 2023 Fall Meeting in San Francisco, CA, USA
6. **10th ASPECT Hackathon**, 2023, Lincoln City, OR, USA
7. **CIDER 2023 Summer Program: Causes and consequences of fluid and magma transport at plate boundaries**, 2023, Berkeley, CA, USA
8. **PyLith Hackathon**, 2023, Golden, CO, USA
9. **3D Modeling of Dynamics in the Hikurangi-Kermadec Subduction System**, Poster presentation at AGU 2022 Fall Meeting in Chicago, IL, USA
10. **3D Modeling of Dynamics in the Hikurangi-Kermadec Subduction System**, Poster presentation at Ada Lovelace Workshop on Numerical Modeling of Mantle and Lithosphere Dynamics, Hévis, Hungary, 2022
11. **3D Modeling of Dynamics in the Hikurangi-Kermadec Subduction System**, Poster presentation at SEG-AGU Workshop, Seattle, WA, USA, 2022
12. **9th ASPECT Hackathon** 2022, Cody, WY, USA
13. **3D Modeling of Dynamics in the Hikurangi-Kermadec Subduction System**, Poster presentation at GeoPRISMs Workshop, Honolulu, HI, USA, 2022
14. **Magma-Assisted Flexure of Hawaiian Lithosphere Inferred From Three-Dimensional Models of Lithospheric Flexure and Active Source Seismic Data**, Poster presentation at AGU 2021 Fall Meeting in New Orleans, LA, USA
15. **8th ASPECT Hackathon** 2021 (virtual)
16. **Using 3D Geodynamic Models of Flexure Backed by Seismic Imagery to Constrain the Rheology and Thermal Structure of the Oceanic Lithosphere Under the Hawaiian Islands**, AGU 2020 Fall Meeting (virtual)
17. **7th ASPECT Hackathon** 2020 (virtual)

VOLUNTEERING

1. **Physics Aid Center Director/Tutor** University of Toronto 2017 – 2019
2. **GSA Earth Sciences Representative** University of Hawaii 2020 – 2021

HONORS, AWARDS & SCHOLARSHIPS

June 2025	CIG Travel Grant to attend the ASPECT Hackathon at Garden City Utah
January 2025	CIG Travel Grant to attend the WorldBuilder Hackathon at GFZ Potsdam, Germany
June 2024	CIG Travel Grant to attend the ASPECT Hackathon at Colorado State University, Mountain Campus
April 2024	Durtsche Award for Research in Geophysics

February 2024	CIG Travel Grant to attend the Worldbuilder Hackathon in Panama City, Florida
December 2023	GSA Travel Grant to attend AGU 2023 Fall meeting in San Francisco, California
September 2023 - February 2024	NSF INTERN Supplemental Funding Award for development of outer rise hydration model using PyLith
July 2023	CIG Travel Grant to attend 2023 ASPECT Hackathon in Lincoln City, Oregon
June 2023	Full funding to attend the 2023 CIDER Summer workshop in Berkeley, California
June 2023	CIG Travel Grant to attend 2023 PyLith Hackathon in Golden, Colorado
December 2022	GSA Travel Grant to attend AGU 2022 Fall meeting in Chicago, Illinois
September 2022	CIG Travel Grant to attend the Ada Lovelace workshop in Heviz, Hungary
July 2022	SEG Travel Grant to attend the SEG-AGU workshop in Seattle, WA
May 2022	CIG Travel Grant to attend the 9th ASPECT Hackathon in Cody, WY
March 2022	GSA Travel Grant to attend the GeoPRISMs workshop in Honolulu, HI
Dec 2020	Stefan Seyb SOEST Endowed Fund
Sept 2019	Graduated with Honours

April 1, 2026